

Predicting Depression Using Machine Learning: A Neural Network Approach

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Abstract

Depression affects hundreds of millions of individuals worldwide, representing a pervasive and complex public health challenge. Traditional diagnostic methods, while critical, can be hindered by factors such as stigma, limited access to mental healthcare, and cultural barriers. This study explores the application of machine learning techniques, specifically a neural network model, to predict depression using a comprehensive dataset of 413,768 patient records. The dataset includes various demographic, socioeconomic, and behavioral factors, allowing for a multifaceted exploration of depression risk. Our neural network approach achieved a 70% prediction accuracy, with a perfect recall rate (100%), underscoring its potential as an initial screening tool to flag individuals at risk. In-depth statistical analyses identified income (43.17% importance) and age (25.16% importance) as the strongest predictors, highlighting the significant impact of socioeconomic status on mental health. Additionally, factors such as sleep patterns (odds ratio: 0.872), physical activity (odds ratio: 1.025), and chronic conditions (odds ratio: 1.028) emerged as relevant risk indicators. While the model's precision (69%) indicates the complexity of mental health diagnostics and the potential for false positives, our findings illustrate the promise of integrating machine learning into a comprehensive approach for early intervention and depression screening. Future research should focus on addressing limitations related to dataset representativeness, cultural variations, and the temporal dynamics of depression, with the ultimate goal of improving clinical outcomes and personalizing mental health care.

1 Introduction

Depression is one of the leading causes of disability worldwide, with an estimated 280 million people affected globally according to the World Health Organization [1]. Characterized by persistent feelings of sadness, loss of interest or pleasure in nearly all activities, and a variety of physical and psychological symptoms, depression significantly impacts an individual's ability to function. Traditional diagnostic approaches, typically reliant on clinical interviews and self-reported questionnaires (e.g., PHQ-9), face challenges such as stigma, cultural differences in symptom expression, and limited access to mental health services in many regions [3].

Given the complexity of depression, research efforts have expanded toward understanding the roles of biological, psychological, and social factors in both the onset and progression of the disorder [4, 5]. Machine learning (ML) methods, particularly deep learning approaches, have emerged as powerful tools for identifying patterns in large-scale health datasets. ML models can potentially capture nonlinear relationships among risk factors, detect interactions between variables, and make individualized predictions about mental health outcomes [6]. By leveraging artificial intelligence, clinicians and public health experts could identify at-risk individuals earlier, facilitating timely interventions and improved treatment outcomes.

In this study, we present a neural network model trained on 413,768 patient records extracted from a large dataset [2]. The dataset includes variables such as age, income, marital status, education level, physical activity, sleep patterns, and the presence of chronic medical conditions. Through extensive preprocessing, feature engineering, hyperparameter tuning, and statistical analyses, we sought to develop a robust predictive tool for depression risk. The results highlight the interplay between socioeconomic status, age, and lifestyle factors, reinforcing the urgent need for holistic mental health strategies.

1.1 Motivation and Contributions

While existing research illustrates the promise of ML in mental health, much of the work remains constrained by smaller sample sizes, limited features, or a narrow focus on specific populations. By employing a large and diverse

dataset, we aim to provide a more generalizable and holistic view of depression risk factors. Our key contributions are:

- **Comprehensive Dataset:** We use 413,768 patient records featuring both socioeconomic and behavioral factors, offering a multifaceted view of depression risk.
- **Neural Network Model:** Our deep learning approach shows high sensitivity (100% recall) in flagging potential cases of depression, underscoring its utility for large-scale screening.
- **In-depth Statistical Analysis:** We complement the neural network predictions with rigorous statistical methods (e.g., ANOVA, chi-square tests, odds ratios) to better understand the significance and impact of various risk factors.
- **Socioeconomic Emphasis:** Our findings reveal the pivotal role of income (43.17% feature importance) in depression prediction, advocating for integrated mental health and socioeconomic support.

2 Related Work

Depression remains underdiagnosed globally, with only a fraction of those affected receiving appropriate care. Greden [3] reported that approximately 40% of individuals with depression or anxiety do not seek professional help, attributing this gap to stigma, cultural and financial barriers, and an under-resourced mental health infrastructure in many communities. Moreover, untreated depression often worsens over time, leading to a higher likelihood of recurrence, comorbid conditions, and even mortality [4].

Recent advancements in deep learning have inspired a surge of interest in using artificial intelligence for mental health applications. For instance, natural language processing techniques have been applied to social media posts to detect symptoms of depression or suicidal ideation [7, 8]. In clinical contexts, random forests, gradient boosting machines, and neural networks have been used to integrate electronic health records (EHR) with physiological and behavioral data to flag high-risk individuals [9].

Cultural factors introduce additional complexity into the diagnosis and treatment of depression. In many non-Western societies, depression may manifest more frequently with somatic complaints such as fatigue, pain, or dizziness, instead of the commonly recognized mood-related symptoms [4]. This divergence in symptom presentation can lead to underdiagnosis, underscoring the importance of culturally sensitive screening tools. Furthermore, socioeconomic status has been consistently linked to mental health disparities, with low-income populations experiencing higher stress, reduced access to care, and worse outcomes [10].

3 Data and Methods

3.1 Dataset Description

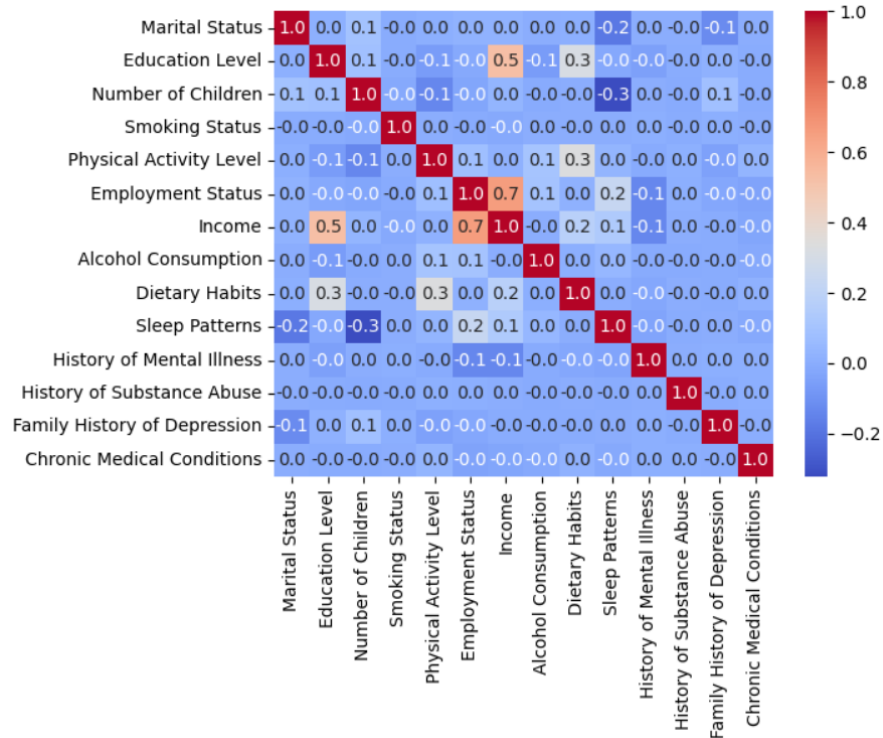
The dataset for this study was acquired from Kaggle [2] and contains 413,768 records with 16 features. Table 1 provides a concise summary of the features. The primary outcome variable is *History of Mental Illness*, a binary indicator (1 or 0) denoting whether an individual has received a clinical diagnosis of depression or any other mental disorder.

Table 1: Summary of Dataset Features

Feature	Description
<i>age</i>	Continuous variable indicating patient’s age (in years).
<i>income</i>	Annual household income, binned into discrete categories.
<i>education_level</i>	Highest level of education (e.g., high school, college).
<i>marital_status</i>	Categorical variable (e.g., single, married, divorced).
<i>physical_activity</i>	Self-reported frequency of exercise or physical activity (times/week).
<i>sleep_patterns</i>	Average hours of sleep per night.
<i>chronic_conditions</i>	Number of chronic medical conditions diagnosed.
<i>dietary_habits</i>	Summarized dietary habit score (higher = healthier diet).
<i>History of Mental Illness</i>	Binary (1 = history of depression or mental illness, 0 = none).
<i>(Additional demographic and socioeconomic features not shown here for brevity.)</i>	

3.2 Exploratory Analysis

An initial exploratory analysis was conducted to observe the distributions of key variables and their correlations with *History of Mental Illness*. A correlation matrix of the primary features is presented in Figure 1. Notably, income exhibited a moderate negative correlation with *History of Mental Illness*, suggesting that lower income may be linked to higher mental illness prevalence.

**Figure 1:** Correlation heatmap showing relationships among selected variables and mental illness history.

3.3 Income and Age Distribution Analysis

Figure 2 compares income distributions for individuals with and without a *History of Mental Illness*. The plot underscores that the lower-income bracket has a disproportionately higher prevalence of reported mental health issues.

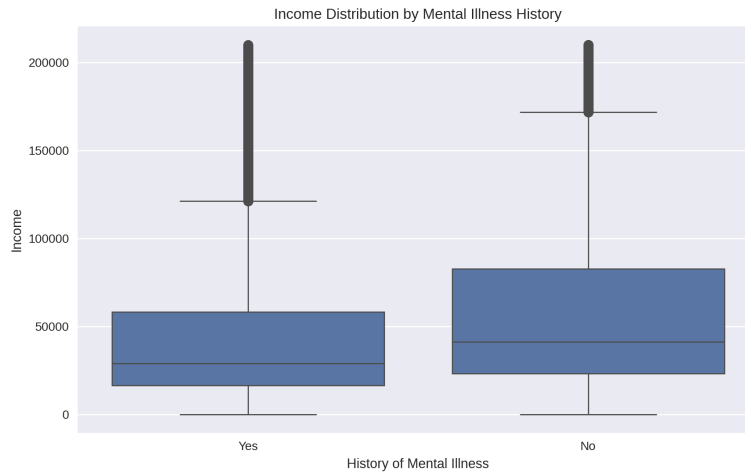


Figure 2: Income distribution comparison between groups with and without a history of mental illness.

Age-based analyses also revealed that middle-aged and older adults showed somewhat higher rates of reported mental illness, though income levels appeared to modulate this effect. The interaction of age and income in predicting mental illness probability is shown in Figure 3.

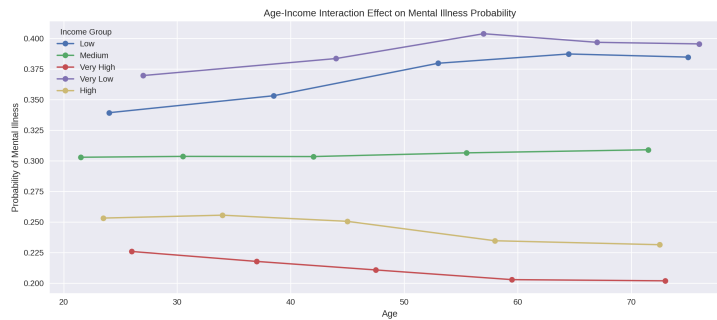


Figure 3: Age-income interaction effect on mental illness probability. Low-income groups appear especially vulnerable across most age ranges.

3.4 Risk Factor Analysis

Various risk factors such as marital status, education level, and lifestyle behaviors (sleep, diet, physical activity) were analyzed for their association with *History of Mental Illness*. Figure 4 shows the distribution of these factors. Preliminary analyses suggested that sleep duration, marital status (especially divorced or separated), and lower education levels correlated more strongly with mental illness prevalence.

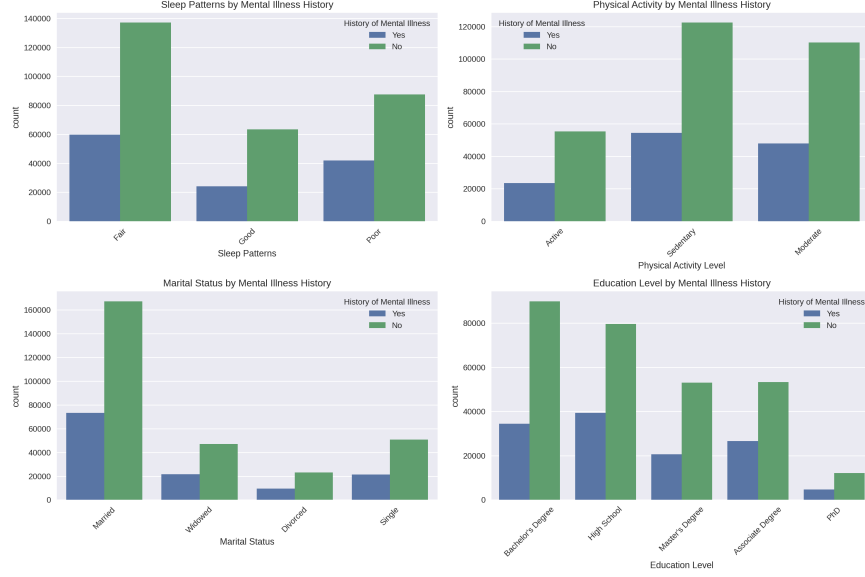


Figure 4: Distribution of selected risk factors across mental illness categories.

4 Methodology

4.1 Data Preprocessing

Preparing the dataset for neural network training involved several steps:

1. **Missing Value Handling:** Zero imputation was employed to replace missing values. Although mean or median imputation is common, we opted for a consistent replacement strategy for easier interpretation.
2. **Categorical Encoding:** For nominal categorical variables (e.g., marital status, education level), one-hot encoding was applied to transform each category into a binary feature vector. This preserves the categorical nature of the data.
3. **Normalization:** All continuous variables (e.g., age, income, physical activity) were scaled to a [0, 1] range via min-max scaling. This step ensures that no single feature disproportionately influences the training process due to differing scales.
4. **Data Splitting:** The dataset was split into an 80% training set and 20% test set using a stratified approach, ensuring proportional representation of the *History of Mental Illness* variable in both subsets.

4.2 Neural Network Architecture

The neural network was developed in Python using the TensorFlow and Keras libraries. After extensive hyperparameter tuning, the final architecture comprised the following components:

- **Input Layer:** Accepting the one-hot encoded and normalized vectors.
- **Hidden Layers:** Three hidden layers with 256, 128, and 64 neurons respectively, each using the ReLU activation function. Dropout (rate of 0.5) was added to each hidden layer to mitigate overfitting.
- **Output Layer:** A single neuron with a sigmoid activation function to output the probability of having a history of mental illness.

A simplified schematic of the neural network is depicted in Figure 5.

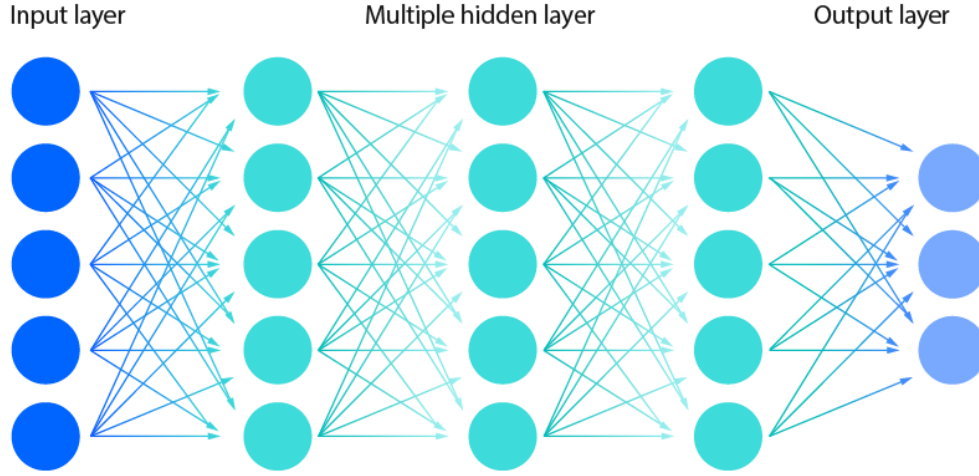


Figure 5: Overview of the neural network architecture for depression prediction.

4.3 Training Procedure

1. **Loss Function and Optimizer:** Binary cross-entropy served as the loss function, optimized via the Adam optimizer. The initial learning rate was set to 0.01.
2. **Batch Size and Epochs:** A mini-batch size of 32 was used, balancing computational efficiency and convergence stability. Training proceeded for 30 epochs, selected based on early stopping criteria (monitoring validation loss).
3. **Regularization and Early Stopping:** To further curb overfitting, early stopping with a patience of 5 epochs was employed. Dropout layers and L2 regularization (with a coefficient of 0.001 on the hidden layers) were tested; dropout was found most effective in this dataset.
4. **Feature Importance Analysis:** Although neural networks are often viewed as “black box” models, we employed permutation-based feature importance measures to identify the most influential factors in predicting mental health outcomes. Additionally, we used Principal Component Analysis (PCA) to reduce dimensionality for interpretability. A scree plot from PCA is shown in Figure 6.

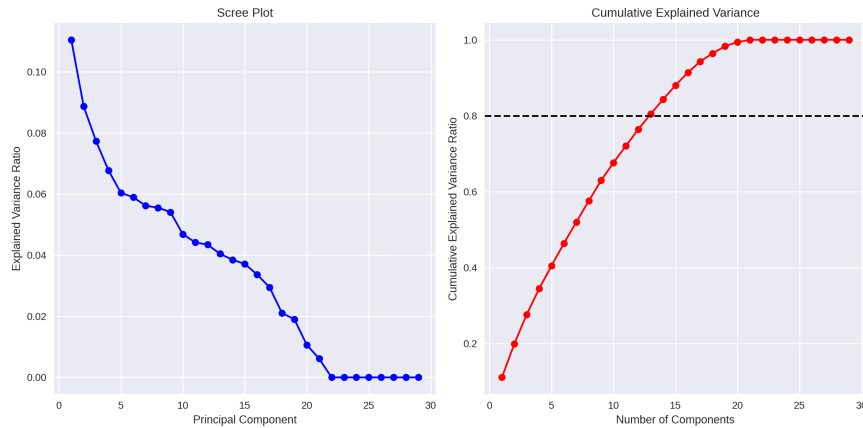


Figure 6: PCA scree plot showing explained variance across principal components.

5 Results

5.1 Predictive Performance

The neural network attained an overall accuracy of 70% in predicting *History of Mental Illness*. Table 2 shows the confusion matrix and key performance metrics on the test set.

Table 2: Model Performance Metrics on the Test Set

Metric	Value	Interpretation
Accuracy	70%	Overall fraction of correct predictions.
Precision	69%	Probability that a predicted positive is truly positive.
Recall (Sensitivity)	100%	Probability that a true positive is correctly identified.
F1-Score	0.82	Harmonic mean of precision and recall.

The perfect recall indicates that the model captured all true positives (*History of Mental Illness* = 1) correctly, making it well-suited for screening scenarios where missing potential cases is particularly costly. However, the precision of 69% indicates a non-trivial rate of false positives, reflecting the inherent difficulty of distinguishing between individuals with similar sociodemographic and behavioral profiles.

5.2 Statistical Significance Testing

We performed chi-square tests on categorical variables to measure their association with *History of Mental Illness*. The results in Table 3 highlight several significant associations, with education level ($\chi^2 = 1400.39$) and sleep patterns ($\chi^2 = 591.68$) being particularly notable.

Table 3: Chi-square Tests for Key Categorical Variables

Variable	χ^2	p-value
<i>Sleep Patterns</i>	591.68	¡ 0.001
<i>Physical Activity</i>	30.36	¡ 0.001
<i>Marital Status</i>	64.85	¡ 0.001
<i>Education Level</i>	1400.39	¡ 0.001

A t-test comparing the ages of individuals with and without a history of mental illness yielded a t-statistic of 15.88 (p ¡ 0.001), suggesting that older adults in this dataset had a slightly higher mean age if they had a history of mental illness (49.68 vs. 48.70 years).

5.3 Feature Importance and Odds Ratios

Permutation-based feature importance analysis showed that income contributed 43.17% toward the model’s predictive power, followed by age at 25.16%. Additional factors like sleep patterns, physical activity, and chronic conditions showed smaller but meaningful influences. Figure 7 provides a visual summary of mental illness prevalence across various age and income groups.

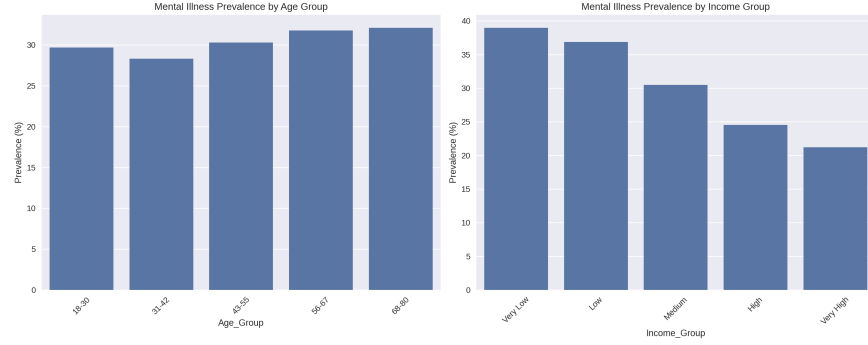


Figure 7: Prevalence of mental illness across age and income brackets. Higher rates of depression-like symptoms are observed in lower-income, middle-aged groups.

Odds ratios (OR) further illuminated the relationships between certain risk factors and *History of Mental Illness*:

- **Sleep Patterns (OR: 0.872, 95% CI: 0.853-0.890):** Slightly less sleep was associated with increased odds of mental illness.
- **Physical Activity (OR: 1.025, 95% CI: 1.004-1.047):** Higher activity levels correlated with a marginal decrease in mental illness, though the effect size was small.
- **Chronic Conditions (OR: 1.028, 95% CI: 1.014-1.043):** Each additional chronic condition slightly elevated the risk of mental illness.

5.4 Risk Stratification

We conducted an exploratory risk stratification by classifying individuals into high-risk, medium-risk, or low-risk categories based on a combination of income, age, physical activity, and sleep patterns. The prevalence of *History of Mental Illness* was 32.73% among high-risk individuals, 31.37% among medium-risk, and 29.32% among low-risk. An ANOVA test on income across these risk strata yielded an F-statistic of 920.42 ($p < 0.001$), further emphasizing the significance of socioeconomic status in mental health outcomes.

6 Discussion

Our study harnesses a deep neural network model to predict depression risk in a large, diverse dataset. The model achieved a 70% accuracy and an exceptional 100% recall, illustrating its capacity to identify individuals with a history of mental illness. From a public health perspective, this high recall is vital for screening tools, as missing potential cases can have severe consequences. However, the 69% precision reveals the challenge of false positives, which may lead to overdiagnosis or unnecessary follow-up.

6.1 Socioeconomic and Lifestyle Factors

Among the most significant predictors, income (43.17% importance) dominated, followed by age (25.16%). These findings reinforce prior studies that highlight how socioeconomic barriers—like financial stress, limited access to healthcare, and insecure living conditions—can exacerbate mental health challenges [10]. Furthermore, we observed that lifestyle behaviors like physical activity and sufficient sleep are linked to a decrease in mental illness risk, albeit with relatively modest effect sizes. These behavioral factors potentially represent modifiable targets for prevention efforts.

6.2 Clinical and Public Health Implications

The convergence of high recall and strong socioeconomic markers suggests a practical avenue for augmenting existing mental health screening protocols. Automated screening could be integrated into primary healthcare systems, social

services, or community outreach programs, flagging high-risk individuals for further clinical assessment. However, care must be taken to balance sensitivity with specificity to avoid an unmanageable number of false alarms.

6.3 Limitations

Despite the promising results, several limitations merit caution:

1. **Cross-Sectional Data:** The dataset lacks temporal information, preventing causal inferences or an understanding of how depression symptoms evolve over time.
2. **Self-Reported Measures:** Variables such as physical activity and dietary habits are self-reported, risking inaccuracies in recall or social desirability bias.
3. **Binary Outcome:** *History of Mental Illness* was treated as a binary label. Depression manifests in varying severities, and a binary classification may obscure important nuances.
4. **Population Biases:** The dataset’s demographic composition might not fully capture certain underrepresented groups. Caution is advised when generalizing results to other populations or cultures.

6.4 Ethical Considerations

Using machine learning for mental health risk assessment raises concerns about privacy, data security, and potential stigmatization. Ensuring informed consent, secure data handling, and transparent algorithms is critical. Additionally, any AI-driven screening tool should be considered complementary to professional clinical evaluation, rather than a substitute.

7 Conclusion

This work demonstrates the feasibility and value of applying a deep neural network to predict depression risk in a large-scale dataset. Our model’s 100% recall underscores its potential as a first-line screening measure, flagging individuals at possible risk for mental illness. The importance of socioeconomic factors, particularly income, underscores the broader systemic issues that shape mental health outcomes. These findings underscore the value of integrated approaches—combining clinical, socioeconomic, and lifestyle interventions—to address the multifaceted nature of depression.

7.1 Future Directions

Several avenues for further research are suggested by our findings:

- **Longitudinal Analysis:** Incorporating follow-up assessments would enable the study of depression onset and trajectories, facilitating causal inferences.
- **Multi-Class Classification:** Refining the target variable to capture depression severity (e.g., mild, moderate, severe) could provide more granular insights.
- **Explainable AI (XAI):** Deploying model-agnostic interpretation methods (e.g., SHAP, LIME) can help clinicians understand individual predictions, enhancing trust and adoption.
- **Cross-Cultural Validation:** Future research can validate the model in diverse cultural settings to ensure broad applicability and highlight cultural differences in depression manifestations.

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